

# What sanctions cost Iranians

## The record since 1979, and three scenarios for 2027

*Four decades of economic pressure have cost ordinary Iranians dearly and squeezed the state's budget too. What that cost bought was leverage, and it was spent only once. Now Iran's oil has stopped altogether. The record suggests what 2027 will bring*

Economic data to 2025, nuclear data to June 2025, events to September 2026

### 13%

How much poorer Iranians were by 2015 than a statistical "twin" built from unsanctioned oil producers

### -16%

Change in household consumption per person, 2017-20

### 0.2 vs 9.9

Tonnes of enriched uranium: the nuclear deal's cap, versus the stockpile in June 2025

### -13%

Projected household consumption per person in 2027 against 2025 if the blockade and the war go on (80% range, -21% to -5%)

Few countries have lived under sanctions as long as Iran. Since the embassy seizure of 1979 America, and at times the United Nations and the European Union, have used economic pressure to change the Islamic Republic's behaviour: first its hostage-taking, then its support for armed groups abroad, and above all its nuclear programme. In 2026 pressure became a blockade. A war that began in February, an American naval blockade and, since August 24th, a sanctions campaign that the Treasury calls Operation Economic Outcast and has billed as "Economic D-Day" have cut Iran's oil exports to nothing.

The debate about whether any of this works is usually conducted in slogans. Hawks say sanctions relief is a cheque written to the Revolutionary Guards. Doves say pressure punishes ordinary people and hardens the regime. Both claims can be tested.

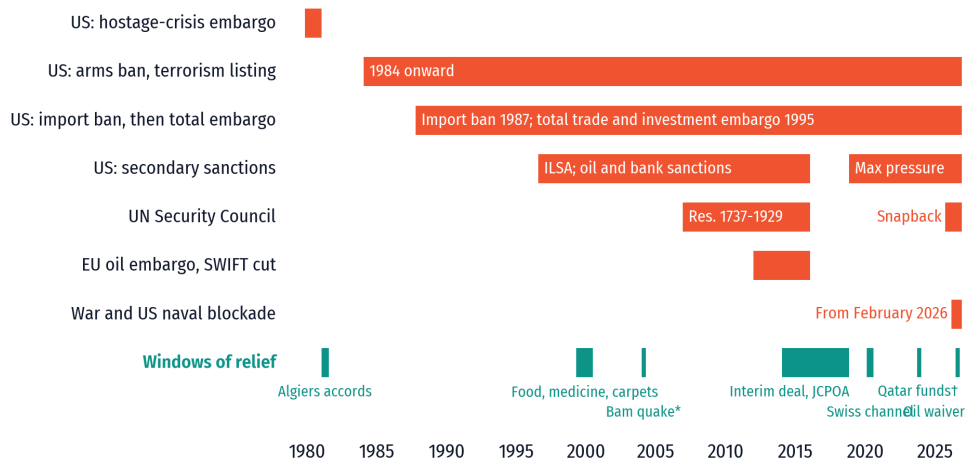
This report sets out what sanctions have cost and what the cost has bought, as a ledger with three columns. The first is ordinary Iranians: their incomes, their prices and what they can afford to consume. The second is the regime's spending priorities, including its military and its allies abroad. The third is the behaviour sanctions were meant to change, measured by the one indicator that is precise, public and independently verified: the International Atomic Energy Agency's count of Iran's enriched uranium. The report then uses the first column, where the evidence is strongest, to project what the blockade will cost households in 2027.

## A record of pressure, with brief interruptions

Sanctions on Iran have never been fully lifted since 1979, with one exception: America unwound its hostage-crisis measures under the Algiers accords of January 1981. Everything since has been partial. In 1999-2000 the Clinton administration allowed sales of food, medicine and aircraft parts and let Americans buy Iranian carpets, pistachios and caviar. After the Bam earthquake of December 2003 the Treasury issued a 90-day licence, but it covered donations through charities, not trade with the state; it is often misremembered as a lifting of sanctions. The interim nuclear deal of 2013 and the Joint Comprehensive Plan of Action (JCPOA), in force from January 2016, were the only large-scale relief. America walked away in May 2018 and reimposed its measures that November.

## Forty-seven years of pressure

Iran: main sanctions regimes and windows of relief, 1979-2026



\*Humanitarian licence for donations via NGOs only. †\$6bn for humanitarian purchases; never spent  
Sources: US Treasury (OFAC); UN Security Council; Council of the EU; press reports

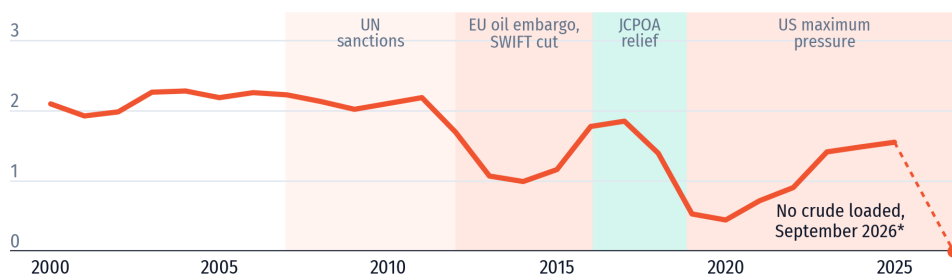
Later windows were narrow or stillborn. A Swiss humanitarian channel opened in 2020. In 2023 \$6bn of Iranian oil money was moved to Qatar for humanitarian purchases and frozen again after the attacks of October 7th, unspent. UN sanctions were reimposed through the "snapback" mechanism in September 2025. The latest window lasted weeks: a memorandum signed in Islamabad in June 2026 brought a 60-day American waiver on Iranian oil sales, and the blockade returned in mid-July.

## The first column: ordinary Iranians

Oil is the channel through which most sanctions bite. Iran's crude exports halved after the EU embargo and the cut-off from the SWIFT payments network in 2012, recovered most of that ground under the JCPOA, and fell by more than 70% between 2017 and 2019. They then climbed back to 1.6m barrels a day by 2025, largely through discounted sales to Chinese refiners via a "shadow fleet", until the blockade of 2026.

### The oil tap

Iran, crude-oil exports, million barrels per day



\*Tanker trackers (Kpler, Vortexa, TankerTrackers), as reported by Bloomberg  
Sources: IMF Regional Economic Outlook via FRED; press reports

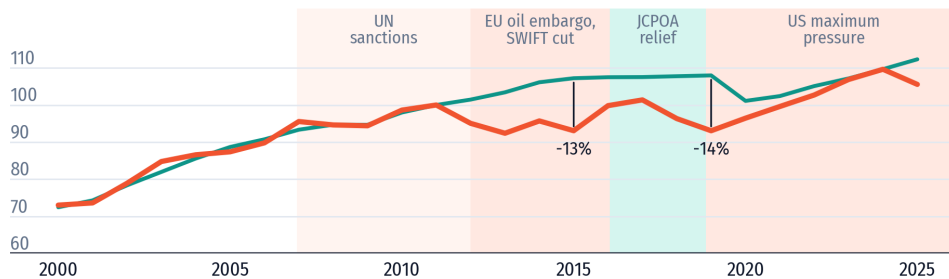
To estimate what this cost, we built a "synthetic Iran": a weighted blend of 29 oil producers that were neither at war nor under heavy sanctions, chosen so that its GDP per person tracked Iran's closely from 1995 to 2011. After the multilateral sanctions of 2012 the two part company. By 2015 Iran was 13% poorer per person than

its twin. Only one of the 29 other countries, put through the same test as if it had been sanctioned, shows a break as sharp. Changing the data source, restricting the twin to net oil exporters or dropping any of its main ingredients gives a gap of 12-15%. Academic studies using the same method put the loss higher, at 17-19%.

## Iran and its twin

GDP per person, 2011=100: Iran and a 'synthetic Iran'\* built from unsanctioned oil producers

— Iran — Synthetic Iran



\*Weighted mix of 29 oil producers matched on Iran's 1995-2011 path; main weights Algeria, Nigeria, Malaysia, Kazakhstan, Cameroon  
Sources: World Bank; Signum Average calculations

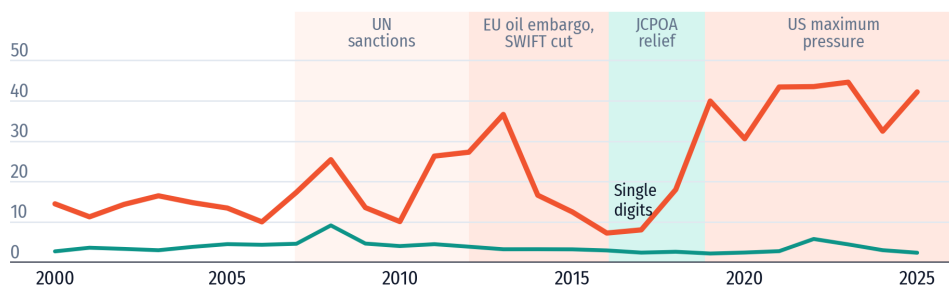
Then, under the JCPOA, Iran's GDP per person rose by 9% in two years while its twin stood still. That is the strongest single piece of evidence that relief mattered. How much of the gap it closed depends on the data: more than half on World Bank figures, all of it on the Maddison Project's, though in the latter case two-fifths of the closing came from the twin's own slump (its largest ingredient there is Nigeria, which was in recession).

The second squeeze looks much like the first. After America's "maximum pressure" campaign began in 2018 Iran's GDP per person fell by 8% in two years, leaving it 14% below its twin. It then recovered, as oil exports found their way to China and the twin was knocked back by the pandemic, and by 2024 the gap had closed. Measured over the whole of 2018-25 the second episode cannot be told apart from chance; its first two years can. Sanctions depress Iran's economy when they stop the oil, and stop depressing it when the oil flows again.

## The price of pressure

Consumer-price inflation, %

— Iran — Peer oil producers, median



Peers: the 26 oil producers in the synthetic Iran with inflation data. Source: World Bank

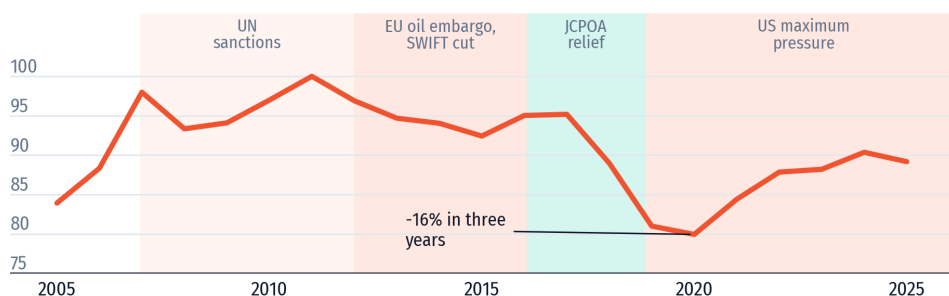
Inflation is where pressure shows most clearly. Iran's inflation was already about 11 points above the median of the same peer countries in 2005-11. The gap widened to 28 points in 2012-13, returned to 11 during the interim deal of 2014-15, narrowed to 5 under the JCPOA and widened to 36 in 2019-25. A simple model of Iran's own history puts the effect of heavy sanctions at about 17 points of extra inflation a year. Two cautions apply. The fall in inflation began in 2014, two years before the JCPOA, when a new government tightened monetary policy and the interim deal steadied the rial; relief was not its only cause. And inflation this

persistent owes much to how Iran's government finances itself when oil money stops.

Inflation falls hardest on those with least: it erodes rial wages, savings and the fixed cash transfer that poorer households rely on. Household consumption per person, a better guide to living standards than GDP, fell by 8% between 2011 and 2015, rose by 3% under the JCPOA and fell by 16% between 2017 and 2020. It then recovered two-thirds of that loss by 2024. In 2025 it was still 11% below its level of 2011. These are national averages. Djavad Salehi-Isfahani, an economist at Virginia Tech who works with Iran's household survey, finds the same shape in the poverty rate: a sharp rise after 2018, then a slow fall, to 16.6% in the year to March 2025.

## Tightening belts

Iran, household consumption per person, 2011=100



Source: World Bank (household final consumption expenditure per capita, constant prices)

A time-series model of household consumption, estimated on Iran's history since 1981, gives a plain answer about the mechanism. In all 60 versions of the model, a halving of real oil revenue cuts the growth of household consumption by three to five percentage points in that year. Whether sanctions are formally in force adds nothing reliable beyond that: the estimate is negative in 57% of versions and statistically significant in none. What hurts households is lost oil revenue, however it is lost, which is why lightly enforced sanctions in 2021-24 hurt far less than the same sanctions strictly enforced in 2019-20. The model's central estimate is that continued relief would have left consumption per person about 11% higher by 2020, with a wide range of 4-19%.

What did the regime buy with relief when it had it? The most visible civilian purchase was aircraft. Iran Air ordered 200 planes after the deal. Only 16 arrived before sanctions returned; Boeing delivered none. GDP grew by almost 9% in 2016, mostly from oil, and unemployment rose from 11.2% to 12.6%. The model above predicted that household consumption would grow by 3.8% and 3.9% in the two JCPOA years; it grew by 2.8% and 0.2%. Relief stabilised prices far more than it created jobs or filled shopping baskets.

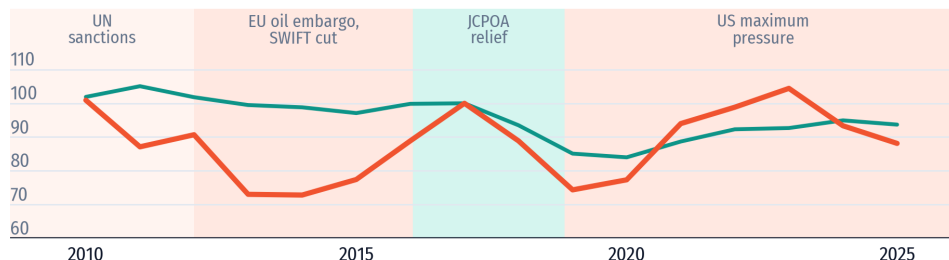
## The second column: the regime's priorities

The hawks' claim is that relief money flows to the Revolutionary Guards and Iran's allies abroad. The official figures give it some support. Iran's recorded military spending, as compiled by SIPRI, a Swedish institute, rose by 29% in real terms between 2015 and 2017. They also show what doves rarely mention: under maximum pressure it fell by 26% in two years, nearly twice as far as household consumption. As a share of government spending the military took 15.5% in 2015, 16.0% in 2017 and 14.6% in 2019. On paper, guns and butter rise and fall together, and guns swing more.

## Guns and butter, on paper

Iran, 2017=100, constant prices

— Official military spending\* — Household consumption per person



\*Excludes off-budget spending: Revolutionary Guards' business income, oil allocated in kind, support to allies  
Sources: SIPRI; World Bank

The official figures miss a great deal. The Guards run a large business empire; since 2019 Iran's budget has allocated oil directly to the armed forces to sell abroad. Support for allies is off the books altogether. A better window is the allies' own payrolls. Hezbollah, Iran's best-funded ally, received about \$700m a year from Tehran according to a 2018 American estimate. It went through two visible cash crunches, and both followed shocks to Iran's income: the oil-price collapse of 2015 and maximum pressure in 2019. In 2019, according to reporting by the Washington Post and the Washington Institute for Near East Policy, married fighters were paid half their salaries, others were furloughed or moved to the reserves, programmes on the group's television station were cancelled and welfare benefits were cut.

So pressure does reach the security apparatus and its clients. What the record does not show is that a poorer Iran became a less dangerous one.

### Why counting rockets does not measure Iran's spending

It is tempting to use the intensity of attacks by Iran's allies as a gauge of Iranian military spending. It fails for three reasons. The biggest campaigns, the strike on Saudi Arabia's Abqaiq plant in 2019 and the Houthis' Red Sea attacks in 2023-24, came when Iran was poorest; they follow political triggers, not budgets. The weapons are cheap: Houthi attack drones cost a few thousand dollars and a larger Shahed-type drone a few tens of thousands, against interceptors costing \$2m or more. And stockpiles built over years break any link between what is fired and what is bought that year.

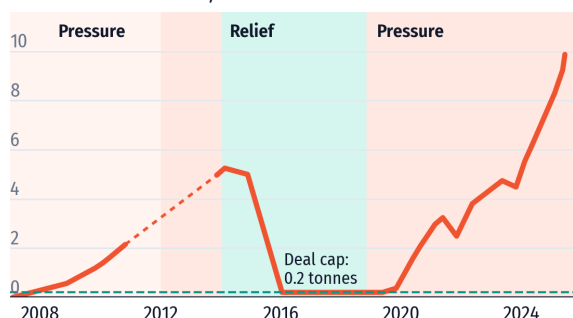
## The third column: the nuclear programme

The IAEA published inspectors' measurements of Iran's enriched uranium every quarter for nearly two decades, which makes this the cleanest series in the ledger. During the sanctions build-up of 2008-14 the stockpile grew by roughly 900kg a year, reaching about 5.3 tonnes by early 2014. The interim deal eliminated Iran's 20%-enriched gas, and the JCPOA cut the stockpile by some 96%, to below its cap of 202.8kg, verified by inspectors until mid-2019.

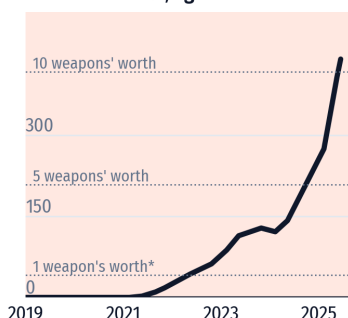
## The nuclear column

Iran's stockpile of enriched uranium, as measured by the IAEA

Total enriched uranium, tonnes



Enriched to 60%, kg



\*IAEA 'significant quantity': 41.7kg of 60% uranium, enough for one weapon if enriched further. Pre-2016 figures converted from uranium hexafluoride; 2008-10 are cumulative production; dashed = no comparable figure. Unverified since June 2025. Source: IAEA

After America left the deal the stockpile grew by about 1,700kg a year, nearly twice the earlier rate. On June 13th 2025, the day Israel's strikes began, the IAEA put it at 9,875kg, including 441kg enriched to 60%, about ten times the 41.7kg the agency regards as enough, if further enriched, for one weapon. Since then the agency has been unable to verify the stockpile's size or whereabouts.

Each camp has a fair objection to the obvious reading. Hawks can say that the pressure of 2012-15 is what brought Iran to the table: the deal that cut the stockpile followed the heaviest multilateral sanctions Iran had faced, and Iran's economy was then 13% smaller than it would otherwise have been. That is a claim about leverage, and the timing supports it. They can add that enrichment to 60% began in April 2021, when enforcement of sanctions was slackening. But the total stockpile grew slightly faster under strict enforcement in 2019-21 (about 2,000kg a year) than under lax enforcement afterwards (1,600kg), so the build-up cannot be blamed on leniency.

Doves, for their part, should not claim that pressure caused the build-up. Iran chose to breach the deal's limits from July 2019, a year after America withdrew, as deliberate retaliation, and better centrifuges would have raised output over time in any policy world. The defensible claim is narrower: pressure without a deal never stopped the stockpile growing, and the only verified cut came when pressure was traded for relief.

## The ledger

Put side by side, the three columns produce a picture that is uncomfortable for both camps.

**Four periods, three columns** Iran under pressure and relief

|                                             | Pressure<br>2012-15   | Relief<br>2016-17          | Pressure<br>2019-25                 | Blockade<br>2026                       |
|---------------------------------------------|-----------------------|----------------------------|-------------------------------------|----------------------------------------|
| <b>Crude oil exports, m b/d</b>             | 1.2                   | 1.8                        | 0.5 (2019-20), then 1.2             | No crude loaded in September           |
| <b>GDP per person vs "twin"</b>             | Down 13% by 2015      | Up 9%; gap at least halved | Down 14% by 2019; level by 2024     | GDP down 10% on a year earlier, spring |
| <b>Inflation above peer median</b>          | +20 pts               | +5 pts                     | +36 pts                             | Inflation 90%                          |
| <b>Household consumption per person</b>     | Down 8% (2011-15)     | Up 3%                      | Down 16% (2017-20), then up 12%     | Down about 5% (our estimate)           |
| <b>Official military spending, % of GDP</b> | 2.4                   | 2.8                        | 2.2                                 | Unknown                                |
| <b>Support to Hezbollah</b>                 | Cuts (2015)           | About \$700m a year        | Fighters on half pay (2019)         | Unknown                                |
| <b>Enriched uranium stockpile</b>           | Rising, ~900kg a year | Capped at 0.2 tonnes       | Rising, ~1,700kg a year; 9.9 tonnes | Unverified                             |

Hawks are right that pressure empties the regime's pockets, including the military's and its allies', and that relief refills them. They are right that the deal followed pressure. They are wrong if they think an emptier pocket changes the nuclear programme by itself: the stockpile grew through both squeezes, and fastest during the second. Doves are right that relief helped ordinary Iranians, chiefly by taming inflation, and that it came with the tightest verified limits the programme has known. They are wrong if they think relief reached households in proportion: consumption rose by 3% while the official military budget rose by 29%.

The most defensible conclusion is narrower than either camp's. Sanctions on Iran work on its economy almost exactly as far as they stop its oil, and the cost lands on households and the state alike. What that cost has bought is leverage, which is worth something only when it is spent. It was spent once, in 2015.

## The year ahead: what the war has done, and what the record implies

The war began on February 28th 2026 with American and Israeli strikes that killed Iran's supreme leader, Ali Khamenei. A ceasefire in April was followed by an American naval blockade; the Islamabad memorandum of June 17th lifted it and waived oil sanctions for 60 days; fighting resumed in July and the blockade returned. On August 24th the Treasury launched Operation Economic Outcast "at President Trump's direction". Mr Trump had used the phrase "Economic D-Day" in the days before, and the Treasury put it in its headline; the treasury secretary, Scott Bessent, called the campaign "an economic D-Day—the single greatest financial offensive ever marshalled against an adversary". It put nearly 60 people, firms and ships under sanctions and extended secondary sanctions, which punish third parties for dealing with Iran, to digital assets, technology, gold, aviation and shipping. Countries that trade with Iran were given a "cure period" to stop.

The measurable damage so far is this. No Iranian crude has crossed the blockade line since mid-July and none was loaded at all in September, according to the tanker trackers Kpler, Vortexa and TankerTrackers; in 2025 Iran exported about 1.6m barrels a day. Oil that was already at sea has kept some money coming in, but that stock fell from about 90m barrels in July to 15m in late September, according to Kpler, and the Wall Street Journal reports that payments may largely dry up by mid-December. Iran's Statistical Centre reports that GDP in the spring quarter was 10.1% lower than a year earlier and the oil and gas sector 26.4% lower. Prices in September were 89.8% higher than a year before, by the Statistical Centre's count; the central bank's figure is 84%. The IAEA says it can provide no information on the size, composition or whereabouts of the uranium stockpile.

## The rial's fall

Rials per US dollar, open market, millions, log scale



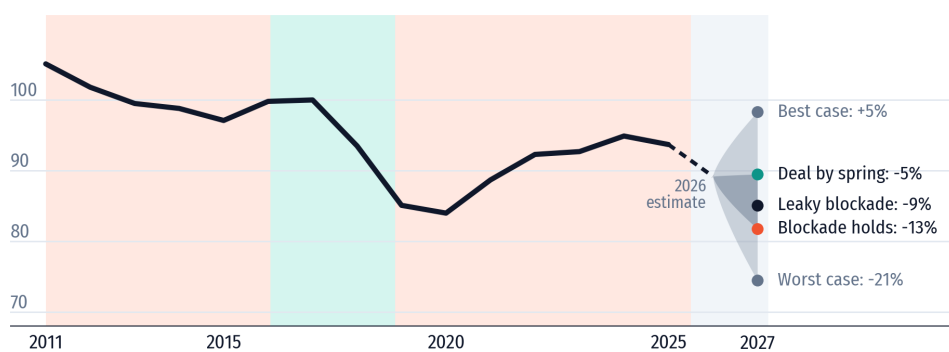
Daily to September 30th 2026. Source: Alanchand

The rial tells the story day by day. A dollar cost 1.35m rials on the open market at the end of 2025, 1.66m on the eve of the war and 1.95m on the eve of Economic Outcast. On September 30th it cost 2.6m. The currency has lost 48% of its value this year, and a quarter of it in the five weeks since the Treasury's announcement.

What happens next depends on a decision that no model can forecast: whether the blockade holds. So we project three scenarios for 2027 and say plainly what the projection rests on. Testing our models against each year since 2008, only one relationship forecast better than a naive guess: household consumption follows oil revenue. For GDP and for inflation nothing beat the naive guess, not even a model fed with the rial's fall, so we offer no model forecast of either. The projection for households uses the IMF's and the Statistical Centre's readings of GDP for 2026 and the oil relationship for 2027. To that it adds a term for war, described below, and the errors the model made in past squeeze years.

## Three ways from here

Iran, household consumption per person, 2017=100, with projections to 2027; labels show change on 2025



Blockade holds: exports about 0.15m barrels a day and war all year. Leaky blockade: 0.65m and war. Deal: 1.2m and war ends in spring  
Dark band: between the central projections; light band: to the 10th and 90th percentiles. Conditional on each scenario, not forecasts of it  
Sources: World Bank; IMF; Statistical Centre of Iran; Signum Average calculations

**Three scenarios for 2027** Household consumption per person against 2025; central projection and 80% range

|                       | Oil exports and war in 2027        | Consumption in 2027 vs 2025 | Inflation after twelve monthst |
|-----------------------|------------------------------------|-----------------------------|--------------------------------|
| <b>Blockade holds</b> | About 0.15m b/d; war all year      | -13% (-21% to -5%)          | 60-100% or more                |
| <b>Leaky blockade</b> | About 0.65m b/d; war all year      | -9% (-17% to -1%)           | 40-65%                         |
| <b>Deal by spring</b> | About 1.2m b/d; war ends in spring | -5% (-13% to +5%)           | 20-35%                         |

If the blockade holds and the war goes on, the central projection is that the average Iranian consumes 13% less in 2027 than in 2025. That would be the lowest level since 2003, and more than a fifth below 2011. About five points of the fall have probably happened already, in 2026. A deal by spring would stop the fall but not reverse it within the year. The deal scenario's 1.2m barrels a day is cautious by the standard of the one precedent: after the deal of 2015 exports rose from 1.2m barrels a day to 1.8m within a year. But then no wells had been shut in and no terminals bombed. The ranges are wide because the model's record is modest, and they overlap: the difference between blockade and deal, about eight points, is better established than the level of either.

**What the war adds**

Oil is not the whole story of 2026. Because the oil price jumped when the fighting began, Iran's oil revenue over the year will fall, on our estimate, by only about a fifth, which by itself would have left household consumption roughly flat. The estimated fall of about 5% is therefore the footprint of the war itself: bombing, blockaded imports, lost jobs.

War enters the projection in two ways. The first is the fighting. In the eight years of the Iran-Iraq war in our data, household consumption grew by about three points a year less than oil revenue alone would explain. We apply that shortfall to each part of 2027 spent at war: all of it in the two blockade scenarios, a quarter in the deal scenario. It deepens the blockade projection by about three points. It is a rough guide, an average over a different war, and too uncertain to pass a statistical test by itself; without it the blockade projection would be -10%, not -13%.

The second is destruction, which outlasts the fighting. The best-documented estimate of physical damage comes from the Foundation for Defense of Democracies, a hawkish think-tank: \$91bn of replacement costs by late April, about half of it military (nuclear sites, missile and drone plants, air bases, ships and air defences). Rystad, a consultancy, put the repair bill for Iran's energy plant alone at \$19bn in April. Iran's government claims \$145bn-270bn, but those figures include lost output. Set against a capital stock of roughly \$2.3trn, the civilian half of the FDD figure is about 2% of everything Iran has built. A standard production function then implies that the economy can make about 1.2% less than before until the damage is repaired, which translates into household consumption about 0.8% lower; even counting the weapons, the loss of capacity is 2.4%. These estimates predate the fighting since July. That loss is already inside our estimate for 2026 and carries forward in every scenario: none of them assumes rebuilding. It is also the smaller part of the 5%. The larger part is disruption, which could reverse quickly with peace. After the ceasefire of 1988 consumption fell again in 1989, then rose by 18% in 1990.

All of this is more likely too mild than too harsh. A total stop in oil exports is a bigger shock than any in the record, so the model is fed the largest fall it has seen (the 66% of 2019) and no more. The two years in which oil revenue last halved, 1986 and 2019, saw consumption fall by 13% and 9% in a single year. And a study of 150 years of wars finds that where a war is fought, GDP is on average 30% lower after five years than it would have been.

On prices, arithmetic is more honest than a model. At September's monthly pace prices would rise by 64% over the next year; at May's pace they would rise by 175%. Once oil payments stop, a government with no other income prints money, which argues for the upper end unless the blockade eases. The one precedent for a way down is 2014, when the interim deal and a tighter central bank more than halved inflation within a year.

The rest of the ledger offers three expectations rather than numbers. The state will be squeezed too: the official military budget fell further than household spending in 2018-19, and it should again. That will not by itself disarm Iran's allies, whose weapons are cheap. And nothing in the record suggests the uranium question will be settled by impoverishment. It was settled once, for a time, by a bargain.

We will return to these projections when the World Bank publishes the figures for 2026 and 2027, and report how they fared.

### What this analysis cannot tell you

It cannot put a reliable dollar figure on how relief money was split between households and the security apparatus, because Iran's military and proxy spending is largely off the books. It cannot establish that pressure caused nuclear escalation, only that it did not prevent it. The household results rest on 45 annual observations and national averages; who gained or lost most requires Iran's household-survey microdata. The projections assume that the relationships of the past 45 years survive a war; the figures for 2026 come from press reports of official and commercial data that we could not check at source.

This report was revised on October 2nd 2026 and corrected on October 5th after a check of its figures against their sources. The economic series were updated to 2025 from the World Bank, the IMF and SIPRI, which changed several results: the fall in household consumption in 2017-20 is 16%, not 15%; GDP did fall behind the twin under maximum pressure, which the first version said it had not; and the section on the regime's priorities no longer says that fighters' pay was protected, which the sources do not support. The section on the year ahead is new. A short version is published on the website under the same title, where the data behind every chart can be downloaded, free to reuse with credit.

## Appendix: data and methods

### Synthetic control

Outcome: log real GDP per person (World Bank, constant 2015 dollars), 1995-2025. Donor pool: 29 oil producers, excluding countries at war (Iraq, Libya, Syria, Yemen) or under heavy sanctions or collapse (Venezuela, Russia); the United Arab Emirates drops out for lack of 2025 data. Weights are non-negative, sum to one and minimise pre-2012 fit error; the largest are Algeria (0.45), Nigeria (0.20) and Malaysia (0.14). Inference uses placebos: the method is rerun on each donor as if it had been sanctioned. Results: Iran 13% below its twin by 2015 (14 log points; placebo  $p=0.07$ ); 14% below in 2019; level in 2024. For the 2018 episode fitted separately,  $p=0.23$  over 2018-25. Checks on the 2015 gap: leaving out each of the five largest donors, 13-15%; net oil exporters only (20 donors), 13%,  $p=0.05$ ; Maddison Project data, 12%,  $p=0.10$ . A backdated test that pretends sanctions began in 2006 finds gaps of at most 4 points, which is the method's noise level.

### Household consumption model

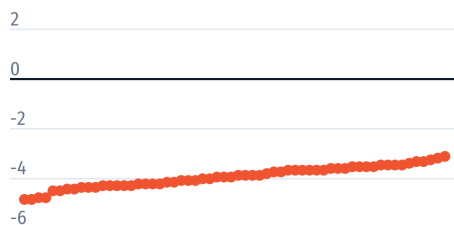
Annual growth of real household consumption per person (World Bank), 1981-2025, regressed on the change in real oil revenue, a date-based index of heavy multilateral sanctions, the US-only embargo era and the Iran-Iraq war, with HAC standard errors. Oil revenue is crude exports (IMF, from 2000; extended back with the growth of production minus domestic consumption) times the Brent price, deflated by US consumer prices. The specification curve re-estimates the model in 60 distinct ways, varying how sanctions are dated, the first year of the sample, and whether 2020, the war years and a lagged term are included. The oil-revenue effect is significant at the 5% level in all 60; the sanctions-in-force effect in none.

## Follow the oil money

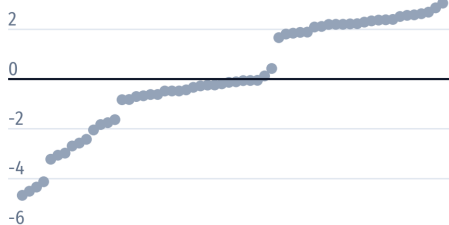
Effect on household consumption growth, percentage points a year, 60 versions of the model, ranked

Significant at 5%    Not significant

Oil revenue halves



Sanctions in force



Versions vary how sanctions are dated, the first year of the sample, and whether 2020, the war years and a lagged term are included  
Source: Signum Average calculations

## Military spending

SIPRI Military Expenditure Database, 1949-2025: spending in constant 2024 dollars, as a share of GDP and as a share of government spending.

## Projections

For 2026, household consumption is estimated from GDP (consumption growth =  $0.43 + 0.64 \times$  growth in GDP per person, fitted on 1981-2025), with GDP growth drawn between the IMF's July 2026 forecast (-5.4%) and the Statistical Centre of Iran's spring reading (-10.1%). For 2027, consumption growth is a constant, plus an elasticity (0.054-0.070) times the change in log real oil revenue, plus a war term times the share of the year at war, in eight versions. Each scenario fixes crude exports, the Brent price and the share of 2027 at war: 0.15m barrels a day at \$90 and war all year; 0.65m at \$85 and war all year; or 1.2m at \$75 and war for a quarter of the year. Exports in 2026 are put at 0.93m barrels a day, from tanker-tracking figures reported in the press. The war term is the average shortfall of consumption growth in the Iran-Iraq war years of the sample: -3.3 points a year, with a standard error of 2.9. Ranges combine model choice, coefficient error and the model's leave-one-out errors; when oil revenue falls by more than 20%, errors are drawn only from past years in which it did. The fall in oil revenue is capped at the largest in the sample (66%, in 2019).

In a rolling test for 2008-25 the oil-revenue model's one-year error was 3.2 points, against 4.3 for a naive forecast. The full model's was 5.7, and no model beat the naive forecast for GDP or for inflation, including one using the rial's annual fall (12.5 points against 10.9). The war term cannot be tested this way, because no year since 2008 was a war year.

Destroyed capital. With output =  $A \times K^a \times L^{1-a}$ , losing a share  $d$  of the capital stock  $K$  lowers potential output by about  $a \times d$ . We take  $a = 0.61$  and a capital-output ratio of 6.3 (Penn World Table, 2023) and GDP of \$363bn (World Bank, 2025), giving a capital stock of about \$2.3trn. The Foundation for Defense of Democracies puts physical damage at \$91bn (April 2026), about half of it military; the civilian half, about \$45bn, is 2.0% of the stock and 1.2% of potential GDP. All \$91bn would give 2.4%, an upper bound because weapons produce no output; Rystad's \$19bn for energy plant alone gives 0.5%. This is arithmetic for scale and is not added to the projection, because the GDP readings for 2026 already contain it.

## Exchange rate

Open-market rials per US dollar, daily from 2008 to September 30th 2026, from the archive published by Alanchand, a price-tracking website. We chart it and use it to date events; we do not republish the series.

## Inflation comparison

Iran's consumer-price inflation (World Bank) minus the median of the 26 donor countries with inflation data: 11 points in 2005-11, 28 in 2012-13, 11 in 2014-15, 5 in 2016-17 and 36 in 2019-25.

## Related research

Gharehgozli (2017, Economics Letters) applies the synthetic control method to the sanctions of 2011-14 and finds that they cut Iran's real GDP by more than 17%. Ghomi (2022, Economics & Politics), with the same method, puts the peak cost at 19% of GDP four years after the sanctions began, and uses household surveys to show that young, rural and less educated households lost most. Laudati and Pesaran (2023, Journal of Applied Econometrics) measure sanctions intensity from newspaper coverage and estimate that sanctions cost Iran one to two percentage points of growth a year. Our estimate of 13% is at the low end of this range. Federle, Meier, Müller, Mutschler and Schularick (2026, American Economic Review) study 150 years of wars and find that in the country where a war is fought GDP is on average 30% lower after five years. Salehi-Isfahani's analysis of the 2024-25 household survey is at [djavadsalehi.com](https://djavadsalehi.com).

## Main sources

World Bank World Development Indicators; SIPRI Military Expenditure Database; Maddison Project and Energy Institute Statistical Review via Our World in Data; IMF World Economic Outlook Update, July 2026, and Regional Economic Outlook (crude exports, via FRED); US Energy Information Administration; Statistical Centre of Iran; Alanchand (exchange rate); Penn World Table; damage estimates from Rystad Energy and the Foundation for Defense of Democracies; tanker-tracking figures from Kpler, Vortexa, TankerTrackers and United Against Nuclear Iran as reported in the press; US Treasury (OFAC) and White House statements on Operation Economic Outcast.

The data behind every chart can be downloaded from the data on GitHub, free to reuse with credit.

## Nuclear data

IAEA quarterly verification reports, 2008-2025. Before 2016 the agency reported uranium hexafluoride, converted here at 0.676kg of uranium per kg; 2008-10 figures are cumulative production and so upper bounds; 2013-14 figures cover the hexafluoride form only. No comparable stock figure exists for 2011-13. The 2016-19 series is shown at the JCPOA cap, which inspectors verified was not exceeded. The series ends on June 13th 2025, the last date for which the agency has an estimate.

## Further sources for this report

IAEA Board of Governors reports, including GOV/2025/24 and GOV/2025/50; Institute for Science and International Security; UN Security Council resolutions; Council of the EU; reporting on Hezbollah finances by the Washington Post and the Washington Institute for Near East Policy (2019); reporting on Iran Air orders and deliveries, and on the costs of drones and interceptors.